Math 190-1: Mathematics of Sustainability **Spring 2026**

Lecture 8: Credibility and Sequential Rationality

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8.1 SequentialGames

Example 8.1 (Sequential Hanging with Friends as Normal Form game) *1. Players: $N = 2$*

2. Strategies:

- $s_1 = \{A, B\}$
- $s_2 = \{aa, ab, ba, bb\}$

3. payoffs:

		Player 2			
		aa	ab	ba	bb
Player 1	A	2,1	2,1	0,0	0,0
	B	0,0	1,2	0,0	1,2

Trying to solve for the pure-strategy Nash Equilibrium, we see

$$\begin{aligned} &\mathrm{BR}_2(A) = \{aa, ab\} \qquad \mathrm{BR}_1(aa) = \{A\}, \quad \mathrm{BR}_1(B) = \{A\} \\ &\mathrm{BR}_2(B) = \{ab, bb\} \qquad \mathrm{BR}_1(fo) = \{\emptyset\}, \quad \mathrm{BR}_1(ff) = \{B\} \end{aligned}$$

This means that the Nash equilbria are: $(A, aa), (A, ab), (B, bb)$.

However, we remember that not all of these are "correct". We now introduce a stronger concept than just Nash Equilibria.

Definition 8.2 *Given a strategy for player i, we say that player i is playing a best response to their opponents in each of their information sets.*

Example 8.3 *Since this is sequential Hanging with Friends, we know that P2 has two information sets, $H_2 = \{\{x_1\}, \{x_2\}\}$, so that means if P2 knows they are at node x_1, then P1 played A, so their best response is a, ie $\mathrm{BR}_2(A) = \{a\}$. Similarly, if we are at x_2, then P2 knows that P1 played B, in which case their best response is b, i.e. $\mathrm{BR}_2(B) = \{b\}$. Now in x_0, P1 believes (correctly) that P2 will play ab, in which case their best response is A. This means that (A, ab) is the unique sequentially rational Nash Equilbrium.*

Remark 8.4 *This precedure is exactly what we did in decision theory, we started from the end and worked our way backward. This is called* backward induction. *We cannot do this if we have information sets that are groups of nodes. (Think about static Hanging with Friends.)*

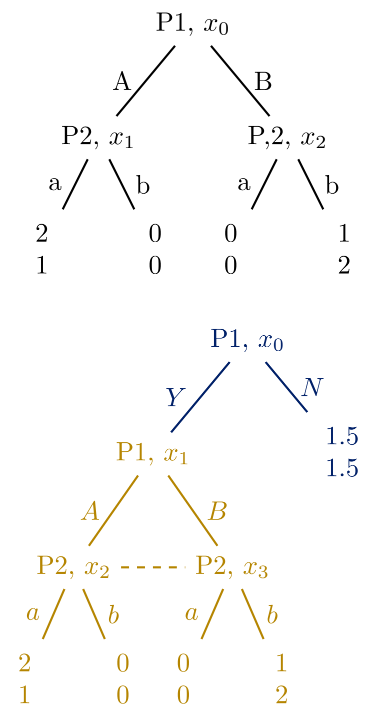


Figure 8.1:

In this game player 1 has four pure strategies. Using the previous convention, the strategy set for player 1 is $S_1 = \{YA, YB, NA, NB\}$, where, for example, YA means that player 1 plans to play Y at x_0 and A at x_1. Note that if player 1 plays either NA or NB then, regardless of player 2's strategy, the game will end after player 1's choice of N.

Player 2, on the other hand, has only two strategies, $S_2 = \{a, b\}$, because they must make their choice without knowing the choice of player 1. Hence the matrix-form representation of this game is as follows:

		Player 2	
		a	b
Player 1	YA	2,1	0,0
	YB	0,0	1,2
	NA	1.5,1.5	1.5,1.5
	NB	1.5,1.5	1.5,1.5

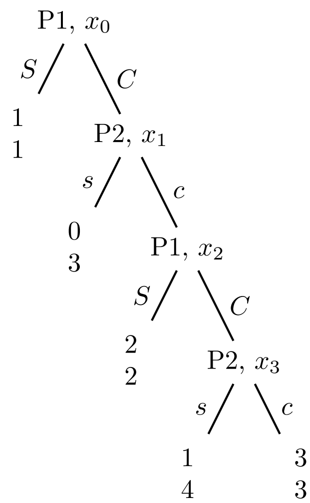
We can find the Nash Equilbrium by looking at best responses

$$\text{BR}_1(a) = \{YA\} \quad \text{BR}_1(b) = \{NA, NB\} \tag{8.1}$$

$$\text{BR}_2(YA) = \{a\}, \quad \text{BR}_2(YB) = \{b\}, \quad \text{BR}_2(NA) = \{a, b\}, \quad \text{BR}_2(NB) = \{a, b\} \tag{8.2}$$

This tells us the Nash equilibria are $(YA, a), (NA, b), (NB, b)$. But we know the game rooted at x_1 is just static Hanging with Friends, so we know that the pure strategy Nash Equilibrium is $(A, a), (B, b)$. So if we are player 1 rooted at x_0, then we have $\text{BR}_1(A, a) = \{Y\}$, and $\text{BR}_1(B, b) = \{N\}$. This means that rational players would only play (YA, a) or (NB, b).

Example 8.5 *We start our backward strategy to see if we can find any subgame- perfect Nash Equilbria.*



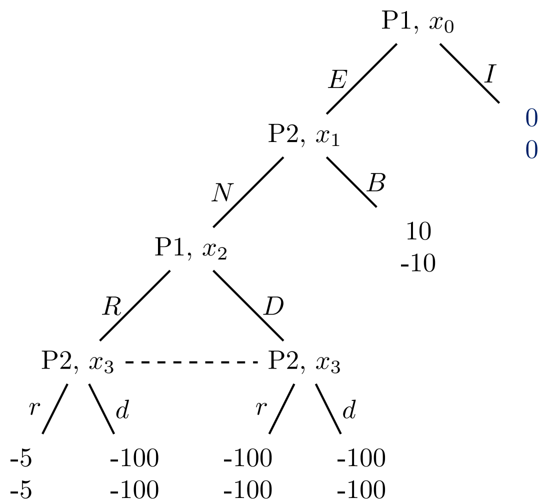
- *If we are at x_3, the best response is to stop since $4 > 3$.*
- *if we are at x_2, the best response is to stop since we know that we will get 1 if they choose to continue C, due to common knowledge and rationality. And they get 2 if they stop so $2 > 1$.*
- *if we are at x_1 the best response is to stop because if we continue c, then they poayoff will be 2.*
- *if we are at x_0 the best response is to stop because $1 > 0$.*

Definition 8.6 *A* subgame *is a subtree of the full game tree (tree nodes and all its successors) such that no information set contains nodes both in and out of the subtree.*

Remark 8.7 *The solution concept for finding Nash Equilbria for (smaller) subgames first and extending these to a larger game, is known as "subgame-perfect" Nash equilbria. This won someone the Nobel Prize in 1994, same year as John Nash.*

Example 8.8 *To analyze this as a game, consider two superpowers, player 1 (the United States) and player 2 (the USSR), that have engaged in a provocative incident. The game starts with player 1's choice to either ignore the incident (I), resulting in maintenance of the status quo with payoffs $(0, 0)$, or escalate the situation (E). Following escalation by player 1, player 2 can back down (B), causing it to lose face and resulting in payoffs of $(10, -10)$, or it can choose to proceed to a nuclear confrontation (N). Upon this choice, the players play a simultaneous-move game in which they can either retreat (R for player 1, r for player 2) or choose Doomsday (D for player 1, d for player 2), in which the world is all but destroyed. If both call things off and retreat then they suffer a small loss due to the mobilization process and payoffs are $(-5, -5)$, while if either party chooses Doomsday then the world destructs and payoffs are $(-100, -100)$.*

		Player 2			
		Br	Bd	Nr	Nd
Player 1	IR	0,0	0,0	0,0	0,0
	ID	0,0	0,0	0,0	0,0
	ER	10,-10	10,-10	-5,-5	-100,-100
	ED	10,-10	10,-10	-100,-100	-100,-100



In this game there are six pure-strategy Nash equilibria, and these are the profiles in the set $E^{Nash} = \{(IR, Nr), (IR, Nd), (ID, Nr), (ID, Nd), (ED, Br), (ED, Bd)\}$.

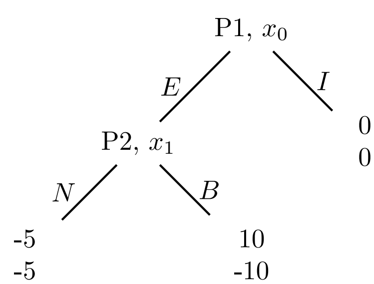
Now to find subgame perfect equilbria

		Player 2	
		r	d
Player 1	R	-5,-5	-100,-100
	D	-100,-100	-100,-100

The two pure strategy Nash Equilbrium are (R, r) *and* (D, d).

Now we look at the subgame rooted at x_1.

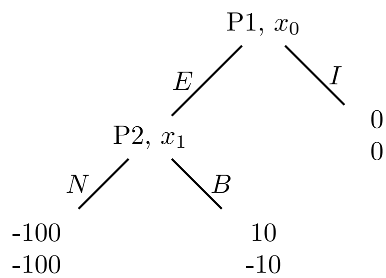
1. *Case #1: Start with NE 1: Player 2 would choose* N *over* B *because losing face is worse than the*



   consequence of mutual defeat.

2. *Case #2: Start with NE 2 Player 2 would choose* B *over* N.

- *if* (R, r) *then* BR_2 *is* N
- *if* (D, d), *then* BR_2 *is* B.



so the subgame perfect Nash equilibria are $(R, Nr), (D, Bd)$, *where* $(s_1(\{x_1\}), s_2(\{x_1\})s_2(\{x_3, x_4\}))$.

Now we consider the whole game

- *if* (R, Nr) *then* $\text{BR}_1 = I$
- *if* (D, Bd) *then* BR_1 *is* E

This means that the 2 subgame perfect pure strategy equilibrium for the game are (IR, Nr) *and* (ED, Bd)

8.2 Multistage Games

		Player 2	
		c	s
Player 1	C	1,1	5,-1
	S	-1,5	4,4

		Player 2	
		l	g
Player 1	L	0,0	-4,-1
	G	-1,-4	-3,-3